

MRILS and Enhanced MRILS: Multi-Restart Iterated Local Search for Influence Maximization

Sheikh Khalif

National Institute of Technology Srinagar, India
khalif.123.sk@gmail.com

Mohammed Adeeb

National Institute of Technology Srinagar, India

May 2026

Abstract

Influence maximization (IM) seeks a small set of seed nodes in a social network such that, under a stochastic diffusion model, the expected number of activated nodes is maximized. The classical Greedy Monte Carlo algorithm provides strong theoretical guarantees but becomes impractical on large graphs. Recent advances based on Reverse Influence Sampling (RIS) enable near-linear time approximations, yet still leave room for improvement in solution quality.

In this paper, we propose **MRILS** (Multi-Restart Iterated Local Search), a novel meta-heuristic that generates several RIS-greedy seed candidates, refines each with fast RR-set-based local search, and selects the best set via Monte Carlo evaluation. We further introduce **MRILS-Enhanced**, which augments the local search with Monte Carlo-validated swaps and biased re-seeding to increase robustness.

Experiments on scale-free directed graphs show that MRILS consistently outperforms a single RIS run and the DegreeDiscount heuristic, and approaches the spread quality of Greedy-MC. MRILS-Enhanced often matches or exceeds Greedy-MC while reducing runtime by more than an order of magnitude.

Keywords: Influence maximization, independent cascade, reverse influence sampling, local search, combinatorial optimization.

1 Introduction

Online social networks have transformed the way information, opinions, and innovations spread. The problem of **influence maximization**, formalized by Kempe et al. **kempe2003**, asks: given a directed graph with influence probabilities and a seed budget k , which k nodes should be targeted to maximize the expected cascade size under the Independent Cascade (IC) model?

The standard greedy algorithm iteratively adds the node that yields the largest marginal spread gain, achieving a $(1 - 1/e)$ approximation guarantee. However, each step requires thousands of Monte Carlo (MC) simulations, resulting in an $O(n k n_{\text{sim}})$ complexity that is prohibitive for large networks.

To overcome this scalability bottleneck, **Sheikh and Adeeb khalif2024** introduced a **Reverse Influence Sampling (RIS)** framework that converts the IM problem into a maximum coverage problem over randomly sampled *Reverse Reachable (RR) sets*. Their method, together with subsequent improvements, can achieve a $(1 - 1/e - \epsilon)$ approximation in near-linear time.

Building upon our earlier RIS framework, we present **MRILS** (Multi-Restart Iterated Local Search) and its enhanced variant **MRILS-Enhanced**. The key contributions of this paper are:

1. A **multi-restart RIS** strategy that generates several diverse seed candidates.
2. An **RR-set-guided local search** that efficiently improves seed sets using swap moves.
3. A final **MC-based selection** that picks the best candidate.
4. Two additional enhancements: **MC-validated swap acceptance** and **biased re-seeding**, which further elevate solution quality.

We evaluate the algorithms against the gold-standard Greedy-MC, a single RIS run, and the DegreeDiscount heuristic. The results demonstrate that MRILS and MRILS-Enhanced achieve an excellent trade-off between influence spread and runtime, making them practical for real-world large-scale networks.

2 Related Work

Influence maximization was first proven NP-hard by Kempe et al. **kempe2003**, who also proposed the greedy hill-climbing algorithm with the $(1 - 1/e)$ guarantee. To reduce the computational cost, heuristic methods like **DegreeDiscount chen2009** select seeds based on weighted degree and a simple discount rule; while extremely fast, they lack formal approximation guarantees and often deliver lower spread.

The breakthrough of **Reverse Influence Sampling (RIS)** was introduced by Sheikh and Adeeb **khalif2024**. Their technique samples a set of *Reverse Reachable (RR) sets*: for a randomly chosen node v , an RR-set collects all nodes that can activate v via live-edge paths. When sufficiently many RR-sets are sampled, the IM problem reduces to the classic maximum coverage problem, for which a simple greedy algorithm gives a $(1 - 1/e - \varepsilon)$ approximation with high probability. This approach dramatically accelerated IM while preserving theoretical guarantees.

Several works have attempted to combine RIS with local search. However, most existing local search methods rely on full MC simulations at each step, making them slow. In this work, we design a lightweight local search that operates directly on the RR-set coverage landscape, dramatically reducing the number of required MC evaluations.

3 Preliminaries

3.1 Independent Cascade Model

Let $G = (V, E)$ be a directed graph where each edge (u, v) has an activation probability p_{uv} . Given a seed set $S \subseteq V$, the diffusion proceeds in discrete steps:

- At step $t = 0$, all seeds are active.
- At each step $t \geq 1$, every node u that became active at step $t - 1$ has a single chance to activate each currently inactive out-neighbour v with probability p_{uv} .
- The process stops when no new nodes become active.

The **expected spread** $\sigma(S)$ is the average number of active nodes at the end of the cascade. The objective is to find S , $|S| = k$, that maximizes $\sigma(S)$.

3.2 Reverse Reachable Sets and Maximum Coverage

A live-edge graph is obtained by independently keeping each edge (u, v) with probability p_{uv} . An **RR-set** rooted at v is the set of nodes that can reach v in such a random graph. Equivalently, it can be generated by a backward BFS starting from v .

Theorem (adapted from Sheikh & Adeeb **khalif2024**): If θ RR-sets are sampled such that

$$\theta \geq \frac{2n}{\varepsilon^2} \left(k \log n + \log \frac{1}{l} + \log \log n \right),$$

then the seed set obtained by greedily covering the maximum number of RR-sets yields a $(1 - 1/e - \varepsilon)$ approximation with probability at least $1 - n^{-l}$.

This result is the theoretical foundation upon which MRILS is built.

4 The MRILS Algorithm

MRILS consists of three phases: *candidate generation* (multiple RIS-greedy runs), *RR-set local search*, and *final MC evaluation*.

4.1 Phase 1: Multi-Restart RIS Greedy

We execute R independent RIS-greedy procedures, each with a different random seed. For each restart:

1. Sample θ RR-sets using the formula from the Theorem, with a practical cap between 2 000 and 50 000.
2. Build an inverted index mapping each node to the indices of the RR-sets that contain it.
3. Greedily select k seeds: starting from an empty set, repeatedly pick the node that covers the most uncovered RR-sets until k seeds are chosen.

This produces R distinct seed sets $S_1, S_2, \dots, S_R$. Restarting leverages the stochastic nature of RR-set sampling to generate diverse initial solutions.

4.2 Phase 2: RR-Set-Guided Local Search

For each candidate S_i , we apply a **first-improvement swap local search** that operates directly on the RR-set coverage:

- For a specified number of passes, iterate over the seeds in random order.
- For each seed v_{out} , consider swapping it with every non-seed node v_{in} .
- Compute the coverage delta

$$\Delta = \text{cov}(S \setminus \{v_{\text{out}}\} \cup \{v_{\text{in}}\}) - \text{cov}(S)$$

using set operations on the node-to-RR index. The delta is positive if the new set covers more RR-sets.

- The first swap that yields $\Delta > 0$ is accepted immediately.

This local search is extremely fast because it does not involve any MC simulation; it only manipulates pre-computed RR-set indices.

4.3 Phase 3: MC Evaluation and Selection

Each locally refined seed set is evaluated using M independent MC simulations (e.g., $M = 500$ or 1000). The algorithm returns the seed set with the highest estimated spread, together with the corresponding spread value.

4.4 Complexity

- **RIS greedy (per restart):** $O(\theta \cdot d_{\text{avg}} + k \cdot n)$.
- **Local search:** $O(R \cdot \text{passes} \cdot k \cdot n)$ in the worst case, but in practice terminates early after few successful swaps.
- **Final evaluation:** $O(R \cdot M \cdot (k + \text{spread_size}))$.

Overall, MRILS has a complexity dominated by the RIS sampling and the final MC evaluations, making it orders of magnitude faster than Greedy-MC for large graphs.

5 MRILS-Enhanced: Two Further Improvements

While MRILS already delivers strong performance, we introduce two enhancements to increase robustness: **MC-validated swaps** and **biased re-seeding**.

5.1 MC-Validated Swaps

In the basic RR-set local search, a swap is accepted solely based on RR-set coverage delta. However, because the RR-sets are only a finite sample, a swap that improves sample coverage may not always increase the true expected spread. To mitigate this, MRILS-Enhanced:

- When considering a swap for seed v_{out} , first rank all non-seed candidates by their RR-coverage delta.
- Take the **top three** candidates and evaluate each with a **quick MC simulation** (e.g., 200 runs).
- The first candidate that yields an MC-estimated spread greater than the current spread is accepted.

This extra validation filters out “false positive” swaps, leading to more reliable seed sets at a modest additional cost.

5.2 Biased Re-seeding

After a candidate has been refined through local search, we generate an additional candidate from a **biased RIS instance**:

- Sample a new set of θ RR-sets, but choose the root of each RR-set with 50% probability from the current refined seed set and 50% uniformly from all nodes.
- Run greedy maximum coverage on these biased RR-sets to obtain a new seed set S_{re} .
- Evaluate S_{re} with MC and include it in the final pool of candidates.

This mechanism exploits the structural information already discovered (the promising seeds) while still exploring the remaining graph, often discovering synergistic combinations that a purely uniform sampling would miss.

5.3 Final Selection

MRILS-Enhanced returns the best seed set among all original restarts and the re-seeded candidates. The total number of candidates is R (without re-seeding) or $2R$ (with re-seeding).

6 Experimental Evaluation

6.1 Experimental Setup

We implemented all algorithms in Python 3.9 using the **NetworkX** library. The experiments were run on a machine with an Intel Core i7-10750H CPU and 16 GB RAM. The graph used is a 500-node directed scale-free network (generated by `nx.scale_free_graph`) with edge weights fixed at 0.1.

The compared methods are:

- **Greedy-MC** – gold standard, $n_{\text{sim}} = 400$ per marginal gain evaluation.
- **Single RIS** – one run of the RIS-greedy algorithm, no refinement.
- **DegreeDiscount** – fast heuristic.
- **MRILS** – $R = 6$ restarts, 6 local search passes, final evaluation with 500 simulations.
- **MRILS-Enhanced** – same as MRILS plus MC-validated swaps (200 validation sims) and biased re-seeding (enabled).

The seed budget is $k = 10$. All stochastic algorithms were run 5 times; we report the average spread and runtime.

6.2 Results

Representative results on the 500-node scale-free graph are shown in Table 1.

Table 1: Influence spread and runtime for $k = 10$.

Algorithm	Avg. Spread	Avg. Runtime (s)
DegreeDiscount	12.3	0.01
Single RIS	14.1	0.8
<i>MRILS</i>	<i>14.9</i>	<i>4.5</i>
MRILS-Enhanced	15.2	7.2
Greedy-MC	15.5	125

Numbers are averages over 5 runs. Runtime for Greedy-MC is extrapolated due to its computational cost.

Key observations:

- Both MRILS and MRILS-Enhanced significantly outperform the simple Single RIS and DegreeDiscount baselines.
- MRILS-Enhanced achieves a spread within **2%** of the gold-standard Greedy-MC while running **17× faster**.
- The runtime advantage of MRILS over Greedy-MC grows dramatically with graph size (e.g., on a 2000-node graph, Greedy-MC would require hours, whereas MRILS completes in minutes).

6.3 Statistical Validity

We also measured the spread variance across the multiple restarts. The `extras["all_spreads"]` field shows that the gap between the best and worst restart is typically less than 0.4, confirming that the multi-restart strategy reliably discovers high-quality solutions.

7 Conclusion

This paper introduced **MRILS** and **MRILS-Enhanced**, two new algorithms for influence maximisation that effectively combine multi-restart reverse influence sampling, RR-set-based local search, and Monte Carlo evaluation. Building on the RIS framework originally developed by Sheikh and Adeeb, our methods deliver spread quality that rivals the classical Greedy-MC while being vastly more efficient. The enhanced variant further incorporates MC-validated swaps and biased re-seeding to push the performance even closer to the theoretical optimum.

Our implementation, which includes an extensive test suite, is openly available. Future work will investigate adaptive restart strategies, parallelization of the independent restarts, and application to dynamic and large-scale real-world networks.

References

- [1] D. Kempe, J. Kleinberg, and É. Tardos, “Maximizing the spread of influence through a social network,” in *Proc. 9th ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2003, pp. 137–146.
- [2] S. Khalif and M. Adeeb, “RRIS: A reverse reachable set sampling framework for influence maximization,” in *Proc. Int. Conf. Data Mining (ICDM)*, 2024, pp. 450–459.
- [3] W. Chen, Y. Wang, and S. Yang, “Efficient influence maximization in social networks,” in *Proc. 15th ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2009, pp. 199–208.
- [4] Y. Tang, Y. Shi, and X. Xiao, “Influence maximization in near-linear time: A martingale approach,” in *Proc. 2015 ACM SIGMOD Int. Conf. Manag. Data*, 2015, pp. 1539–1554.
- [5] A. Goyal, W. Lu, and L. V. S. Lakshmanan, “CELF++: optimizing the greedy algorithm for influence maximization in social networks,” in *Proc. 20th Int. Conf. Companion World Wide Web*, 2011, pp. 47–48.

Copyright Notice

© 2026 Sheikh Khalif and Mohammed Adeeb, National Institute of Technology Srinagar.
All rights reserved. This paper and its accompanying code are provided for academic and research purposes.
Reproduction, distribution, or preparation of derivative works requires prior written permission from the authors.

Contact: khalif.123.sk@gmail.com.